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On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.

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arindam-codes Update weekSchedule.md 4f4ac25 · last month

README.md	Update roadmap with semester dates an...	last month
weekSchedule.md	Update weekSchedule.md	last month

README

🚀 Career Strategy Roadmap (MIT-Based Learning Path)

🧠 Year 1: Foundations (Current Semester) (April, 2026 - July, 2026)

MIT Course	Course Name	What You Learn	Track 1 (Game/Systems)	Track 3 (AI/Quant/Robotics)
6.1010	Fundamentals of Programming	Software design	Roblox architecture patterns	Systems thinking
18.01	Calculus	Derivatives, integrals	Player growth curves	ML math
6.1200	Discrete Math	Logic, proofs, combinatorics	Game state representation	Algorithm design
8.01	Physics	Mechanics	Better game physics	Simulation foundations
6.1020	Software Construction	Software Const	Unknown	Unknown

⚙️ Year 1: Second Semester (August, 2026 - November, 2026 + December, 2026)

MIT Course	Course Name	What You Learn	Track 1 (Game/Systems)	Track 3 (AI/Quant/Robotics)
6.1060	Software Performance Engineering	Software Performance	Unknown	Unknown
18.02	Multivariable Calculus	Vectors, gradients	3D game math	ML optimization
6.1210	Algorithms	Sorting, searching, graphs	Efficient game logic	Anti-cheat algorithms
8.02	Physics II	Electricity, magnetism	Hardware foundations	Robotics sensors
6.1910	Computation Structures	Computer architecture	Performance optimization	Hardware understanding

🖥️ Year 2: Core Systems (April, 2027 - July, 2027) + (August, 2027 - November, 2027 + December, 2027)

MIT Course	Course Name	What You Learn	Track 1 (Game/Systems)	Track 3 (AI/Quant/Robotics)
6.5060	Algorithm Engineering	Unknown	Unknown	Unknown
6.1800	Computer Systems Engineering	Operating systems	Server-side Roblox	Distributed systems
6.2000	Electrical Circuits	Circuit analysis	Gaming hardware	Robotics hardware
6.2510	Introduction to Machine Learning	Basic ML algorithms	Better anti-cheat	ML foundations
18.03	Differential Equations	ODEs	Physics simulation	Control systems
6.1310	Power Electronics	Power systems	Hardware projects	Robotics power
18.06	Linear Algebra	Linear Algebra	Unknown	Unknown
18.05	Probability stats	Prob Stats	Unknown	Unknown
6.1220	Algorithms	Algos	Unknown	Unknown

🌐 Year 3: Depth Areas (April, 2028 - July, 2028) + (August, 2028 - November, 2028 + December, 2028)

MIT Course	Course Name	What You Learn	Track 1 (Game/Systems)	Track 3 (AI/Quant/Robotics)
6.4110	Distributed Systems	Scalable systems	Roblox backend at scale	Metaverse infrastructure
6.4210	Robotics	Kinematics, control	Game physics	Robotics engineering
6.4310	Probability	Random processes	ML anti-cheat	Quant trading
6.4400	Computer Graphics	Rendering pipelines	Game engine customization	Graphics programming
6.7900	Machine Learning	Deep learning	AI game systems	AGI foundations
6.8010	Signal Processing	Filters, transforms	Audio generation	Sensor processing
6.3100	Dynamic Programming	D.P.	Unknown	Unknown
6.4212	Reinforcement Learn	RL algorithms	NPCs that learn	AGI systems
6.8300	Computer Vision	Image recognition	Mocap from video	Robotics vision

🤖 Year 4: AI & Advanced Systems (April, 2029 - July, 2029) + (August, 2029 - November, 2029 + December, 2029)

MIT Course	Course Name	What You Learn	Track 1 (Game/Systems)	Track 3 (AI/Quant/Robotics)
6.8611	Quant Methods	Q.M.	Unknown	Unknown
6.8200	Sensorimotor Learning	Learning for control	Game AI	Robotics integration
6.8510	Advanced Data Structures	Cutting-edge DS	Ultra-efficient systems	Research-level work
6.8810	Optimization	Convex, non-convex	Trading systems	Engineering design
6.8350	Intelligent Multimodal Interfaces	HCI + AI	JARVIS-like interfaces	Tony Stark UI
6.8701	Systems for AI	Systems	Unknown	Unknown
6.3950	SERC	SERC	Unknown	Unknown

Releases

No releases published
Create a new release

Contributors 1

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